Sony ID7000 Standard Operating Procedure & User Guide
UC Davis Flow Cytometry Shared Resource

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Computer Login Details:

Account: FCM  Password: FCM

**Part 1: System Startup**

1. Turn on the ID7000. (Power button on front)
2. Check DI Water level (Sheath).
    a. If the DI Water tank is not full:
        i. Remove the lid from the DI Water tank and place into the sheath lid holder to keep the line clean.
        ii. Take the Water tank to the sink and fill it with the house DI water faucet (left most). *Note: the spout lever leaks a bit when actuated, take care to minimize spills
        iii. Return the Water tank to the ID7000 and reattach the lid.
        iv. Make a note to email JVD as the Water tank should have been filled at the end of every run if <50% full.
    b. If DI Water tank is full, move to the next step.
3. Check Waste tank level.
    a. If the Waste tank is >50% full:
        i. Remove the lid from the Waste tank and place into the waste lid holder.
        ii. Take the Waste tank to the sink and dump it down the drain.
        iii. Refill the Waste tank with 1L of 100% bleach (under the sink).
        iv. Return the Waste tank to the ID7000 and reattach the lid.
        v. Make a note to email JVD as the Waste tank should have been emptied at the end of every run if >50% full.
    b. If Waste level is <50% move to the next step *If you have a large experiment, you should start with an empty Waste tank (1L bleach only) follow step 3a.
4. Open the ID7000 software.
5. Log into the ID7000 software
    a. Account name: “PI last name (capital first letter)” + “Lab”. E.g. SmithLab
    b. Password: “PI Last name” + “Current year or previous year”. E.g Smith2024
    c. ***Note: The ID7000 forces users to change their password every year(365days). When this happens, please update the password on your lab’s account to “PI last name (capital first letter)” + “Current year”.***
6. Allow the system to connect.
7. Wait for System Prime window to popup.
8. Click *Start* in the System Prime window.
    a. This will take about 5min to complete.

**Part 2: Optical Alignment and Performance QC**

1. Obtain *Sony Beads Control* box from the fridge.
2. Optical Alignment (If the *Optical Alignment* wizard window pops up do the following, otherwise move to Daily QC):
    a. **First** perform contrad cleaning procedure for the system:
        i. Click the *Cytometer* tab at the top.
        ii. Click the "Decontamination" button.
        iii. Select the "Bleach Cleaning and Rinse" from the drop down.
        iv. Follow the cleaning wizard with the following adjustments:
            1. Bleach tube: Use a **20% Contrad** solution instead
                a. Fill fresh FACS tube to weld line with 100% contrad then fill ⅔ with DI water.
            2. Rinse tube: Use 100% Di Water
    b. **Then follow the wizard prompts with the notes below**
        i. Procedure uses **AlignCheck** beads.
        ii. To prepare the **AlignCheck** bead control (do this every time):
            1. Carefully pipette 450uL of fresh DI water into a fresh tube.
            2. Vortex **AlignCheck** bead stock right-side up, upside down and sideways.
            3. Carefully add 2 drops of the AlignCheck beads to the tube.
            4. Label and Vortex the tube.
        iii. Make sure to check the bead lot of the AlignCheck beads with the drop down in wizard.
3. Daily QC:
    a. If >5 days since last QC was run, perform a contrad cleaning procedure.
        i. Click the *Cytometer* tab at the top.
        ii. Click the "Decontamination" button.
        iii. Select the "Bleach Cleaning and Rinse" from the drop down.
        iv. Follow the cleaning wizard with the following adjustments:
            1. Bleach tube: Use a **20% Contrad** solution instead
                a. Fill fresh FACS tube to weld line with 100% contrad then fill ⅔ with DI water.
            2. Rinse tube: Use 100% Di Water
    b. If <5 days since last QC was run or after above contrade cleaning, run the Daily QC
        i. Click the *QC* tab on the left side of the screen.
        ii. Click on *Daily QC* option.
        iii. In the *Status* box date field note the last time the Daily QC was ran.
        iv. If the last run date is >24hr run a Daily QC:
            1. Click "Start Daily QC", **follow the wizard with the notes below**:
                a. Procedure uses **AlignCheck** beads.
                b. To prepare the **AlignCheck** bead control (do this every time):
                    i. Carefully pipette 450uL of fresh DI water into a fresh tube.
                        1. *System calibrates sample flowrate with Daily QC. It is very important to get the bead concentration correct.*

ii. Vortex **AlignCheck** bead stock right-side up, upside down and sideways.
iii. Carefully add 2 drops of the AlignCheck beads to the tube.
iv. Label and Vortex the tube.

c. Make sure to check the bead lot of the AlignCheck beads with the drop down in wizard.

c. Click the latest Daily QC run and confirm the result.

1. If the QC passed, move onto the next step.
2. If the QC fails or has a warning, identify the warning, notify Flow Core Personnel, and run a Contrad cleaning procedure for the system.
    a. Click the *Cytometer* tab at the top.
    b. Click the "Decontamination" button.
    c. Select the "Bleach Cleaning and Rinse" from the drop down.
    d. Follow the cleaning wizard with the following adjustments:
        i. Bleach tube: Use a **20% Contrad** solution instead
            1. Fill fresh FACS tube to weld line with 100% contrad then fill ⅔ with DI water.
        ii. Rinse tube: Use 100% Di Water

4. System is QCed and ready to run.

**Part 3: Compensation/unmixing using the Reference Library (you will setup a separate compensation experiment)**

- Compensation controls need to be run at least once for each fluorochrome you plan to use.
- Tandem dye fluorochrome compensation controls need to be ran for **each new product or lot of the same product.**
- Tandem dye fluorochrome products should have their **reference spectra updated/refreshed every month**.
- Good quality spectra that almost saturate (>10^5) the primary detector for a given fluorochrome will give the best results (adjusted with primary laser's detector array gain).
- Bring a printout with details about each fluorescent reagent in your panel:
    - Fluorochrome name
    - Product lot
    - Antibody clone (If applicable)
    - Relative intensity of fluorochrome,
        - Flurofinder is a great place to find this information.
            - PE for example is 5/5 while FITC is 2/5

***Setting up the compensation experiment***

1. Click on the *Experiment* tab.
2. Click on the Experiment *Designer* option.
3. Turn ALL lasers on by clicking the *355nm, 405nm, 488nm, 561nm, and 637nm* check boxes (bottom middle).
4. Using the *Add New Spectral Reference* box:
    a. Type the name of each fluorochrome in your assay (you will need a single-color control for marker in your assay), select it from the filtered list, click "Add".
    b. If your fluorochrome is not in the list:

i. Add a custom fluorochrome by fully typing the name of your Fluorochrome into the box and then click “Add”.
ii. The use custom popup window set the *Start* detector to the detector that captures the peak emission of the fluorochrome, then set the *End* detector as Start+1.

5. Add reagent and marker information:
    a. Label: This should be something to help differentiate each set of markers (IE panel name)
    b. Marker name (IE CD3)
    c. Manufacturer
    d. Product lot
    e. Antibody clone (If applicable)
6. Click “Next”
7. Choose the plate type.
    a. Deselect the “Use a common unstained control for single stained controls” check box.
        i. This generates positive and negative gates per control.
        ii. If you need a separate unstained control you can add manually during compensation calculation. Make sure to record unstained control!
    b. 24-well tube rack/96-well plate (std/half deep/deep)/384 well-plate
8. Click “Next”
9. Confirm *Mode: Standardization* If it is not, please reach out to flow core staff for assistance.
10. Set the *Acquisition Settings*
    a. Cleaning Mode: Inner and Outer
    b. Acquisition Offset Time: 3-4sec
    c. Select “Enable low dead volume mode” – **Optional (5-10uL)**
    d. Select “Enable sample agitation”.
    e. Pattern: Once
    f. Mode: Normal
11. Click “Next”
12. Select the order for acquisition to your preference, “Left-To-Right” is recommended.
13. Click “Next”
14. Enter a name for your experiment. (Recommended Name: *Panel Name – Compensation*)
15. Click “Next”
16. Review experiment details and layout.
    a. If you are using tubes, vortex and load each single-color control into its proper position in the tube rack.
    b. If you are using a plate, ensure the sample (software) layout matches your plate layout, if it does not customize the sample (software) layout to match your plate.
17. Click “Create Experiment”

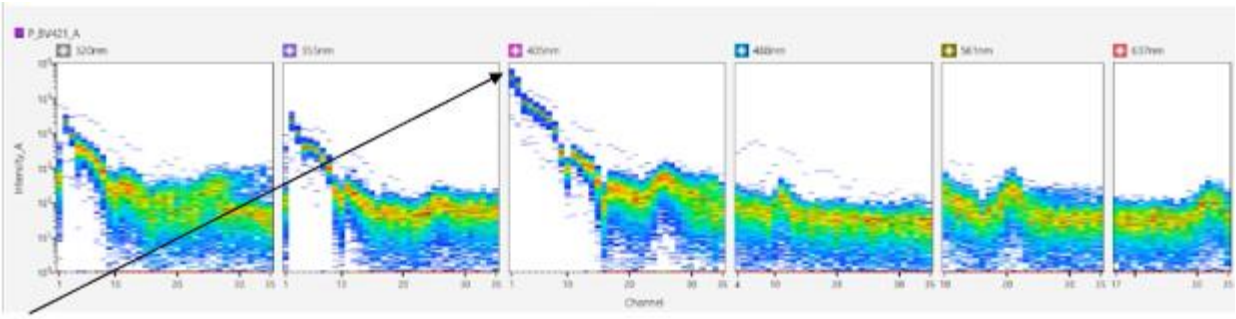
***Setting the gains for FSC, SSC and each detector array to maximize sensitivity.***

- In this section you will need to identify the brightest fluorochromes off each laser in your panel. Fluorochromes that have the same or an unknown level of intensity will all need to be ran to determine who is the brightest.
    - Fluorofinder is a great free resource to identify fluorochrome intensity and peak emission.
        - https://fluorofinder.com/
    - Common brightest fluorochromes include:

- UV: BUV (615, 661, 737)
- Violet: BV421, SB 436
- Blue: BB515, BB700
- Yellow-Green: PE, PE-CF594, PE-Dazzle594, PE-Cy7, PE-Fire700, PE-Fire810
- Red: APC, AF647, R718, APC-Fire810

18. Load the tube rack or plate into the ID7000. (A1 should be forward left side facing you)
19. Click the "Load Tube Rack/Plate" button. 
20. Highlight the tube/well for the control you would like to check (Should be brightest fluorochrome on its laser).
21. Right click the tube/well and select "Set current position" (Selected well/tube denoted with black arrow).
22. Click on the FSCXSSC plot.
23. Click the *Y-Axis* drop down (top of screen) and select "Log".
24. Open the *Instrument Settings* window by clicking the "Instrument" button. 

25. In the *Instrument Settings* window:
    a. Click the *Flow Control* tab.
    b. Set flow rate to "1".
    c. Click the *Detector & Threshold* tab.
26. Click the "Preview" button to begin acquiring the sample. 
27. Monitoring the FCSXSSC Plot:
    a. Use the Detector & Threshold tab (*Instruments Settings* window) adjust the FSC and SSC gains to bring your single-color control population/s on scale and above the threshold.
        i. Recommended settings for beads:
            1. FSC Gain: ~8 (Plot position: 0.1-0.3)
            2. SSC Gain: ~3.78 (Plot position 10^3-10^5)
    b. Move the "A" gate over the singlet population.
28. In the Histogram plot:
    a. Adjust the positive interval/linear gate.
    b. Adjust the negative interval/linear gate. (only present when universal negative not used)
        i. Adjust axis scaling as needed to ensure all negative events are visible and clustered.
29. Monitoring both Spectral Ribbon plots (All Events and Histogram gated):
    a. Use the Detector & Threshold tab (*Instruments Settings* window) adjust the gain for the proper laser's detector array to ensure the events in the primary detector are at 10^5 or above but remain on scale (<10^6). *See example of BV421 below.
        i. Generally, you will want to use a gain between 4.5-5.5.
        ii. If you do not have a fluorochrome for every laser, set those laser/s gain to 4.8 or match the primary laser's gain which ever is less.
30. Click the "Preview" button to stop the uptake of sample.



BV421 is on scale for primary detector and signal is >10^5 but <10^6.

31. Repeat steps 20-29 for each remaining “brightest fluorochrome on its laser” control.
    a. As you work through your brightest fluorochromes you may find that if very high gains are needed on one or more lasers this can cause some markers to become “brighter” on other lasers than their primary. This is a function of gain and not indicative of the true spectral profile of your marker. Try to reduce the gain on the non-primary laser taking care not to allow any of the “brightest fluorochrome on its laser” controls to drop below 10^5 in its primary detector.
32. In the *Instrument Settings* window, click “Close”.

***Recording the single-color controls***

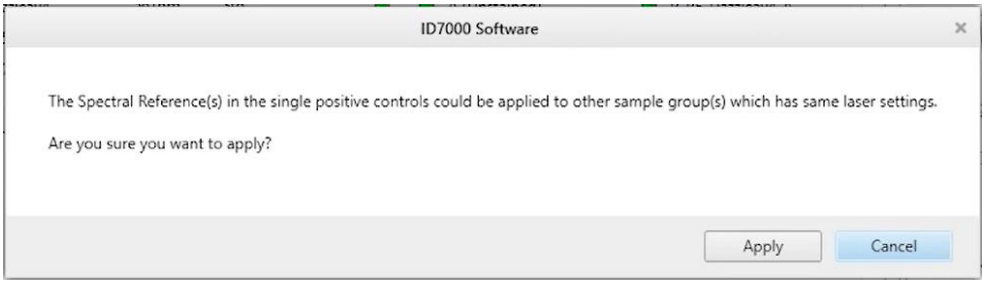
33. Multi-select all control tubes/wells.
34. Right click and select “Add to Auto Acquisition Group”.
    a. Note: This may be greyed out if tubes/wells were automatically placed in *Auto Acquisition Group*.
35. Click the “Stopping Condition” button 
36. In the *Stopping Condition* window
    a. Set the events to record to 10k-25k.
    b. Click “Close”.
37. Click “Auto Acquire” button  to begin recording your tube rack/plate.

***Generate unmixing matrix***

38. For each single-color control:
    a. Select the tube/well with data.
    b. Confirm the positive and negative gates in the Histogram plot.
39. Select the *Unmixing* tab at the top.
40. Click the “Unmixing Settings” button.
41. In the Unmixing Settings window check that the positive and negative gates are all set properly
    a. Gates for positive should be on the highest prevalence, brightest, singlet population within your control. Sometimes this can be quite smeary, esp in the case of cell controls. In these cases, try to use the brightest single cell events.
    b. The negative gate should be set on a negative matching particle or cell type.
        i. Matched beads (product and lot) or matched cell type as the positive population.
42. Deselect the “Use as AF” check box in the Autofluorescence subtraction section.

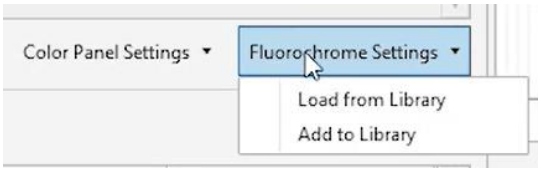
a. This is a feature you can take advantage of when running a real experiment on your experiment cells.

43. Click the “Calculate” button to calculate the matrix.
44. If there are any saturating events, click the “Calculate Spectral Reference without Saturated event/s” checkbox and click “Continue” in the popup.
45. Click “Apply” in the popup




***Adding fluorochrome spectra to Reference Library***

46. In the bottom right of the “Unmixing Settings” window:
    a. click the “Fluorochrome Settings” button.
    b. In the drop down select “Add to Library”



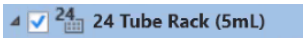

47. In the “Add to Reference Library” window:
    a. Make sure all fluorochrome spectra you wish to add are checked.
    b. Add or update labels as needed.
    c. Click the “Add to Reference Library” button.
48. Click the “Unload” button and remove the tube rack/plate. 

**Part 4: Setting up and Running the Experiment**

- This SOP is expecting that you will have all reference controls already in the Reference Library, if you have new colors or need to refresh a 1 or more controls in the Reference Library, please follow the previous section with just those markers and then return to this point.

***Setting up the acquisition experiment***

1. Click the *Experiment* tab on the left.
2. Click on the *Experiment* tab.
3. Click on the Experiment *Designer* option.
4. Turn ALL lasers on by clicking each laser’s check box in the bottom left corner.
5. Add each fluorochrome’s reference spectra from the Reference Library (top section of window) in your assay to the Selected Spectral references box:
    a. “CTRL” clicks each reference spectra for each fluorochrome (<u>you must have a reference for each fluorochrome in your assay</u>) in your assay.
    b. Click the “Select” button to add the selected reference spectrums to your experiment.

6. Click “Next”
7. Choose the plate type.
    a. 24-well tube rack/96-well plate (std/half deep/deep)/384 well-plate
8. Click “Next”
9. Confirm *Mode: Standardization* If it is not, please reach out to Jonathan Van Dyke for assistance.
10. Set the *Acquisition Settings*
    a. Cleaning Mode: Inner and Outer
    b. Acquisition Offset Time: 3-4sec.
    c. Select “Enable low dead volume mode” – **Optional (5-10uL)**
    d. Select “Enable sample agitation”.
    e. Pattern: Once
    f. Mode: Normal
11. Click “Next”
12. Select the order for acquisition to your preference, “Left-To-Right” is recommended.
13. Click “Next”
14. Enter a name for your experiment. (Recommended Name: *Panel Name – Compensation*)
15. Click “Next”
16. Review experiment details.
17. Click “Create Experiment”
18. Select the first tube rack or plate in the Experiment Layout. 

19. Click tube/well “A1” from the tube rack/plate map.
20. Click on the FSCXSSC plot.
21. Click the *Y-Axis* drop down (top of screen) and select “Log”.
22. Label tubes/wells in software:
    a. Click the “Map” button, upper right side of plate map. 
    b. Edit plate map to label each well.
    c. If using the tube rack, load your tubes into the rack according to the map.

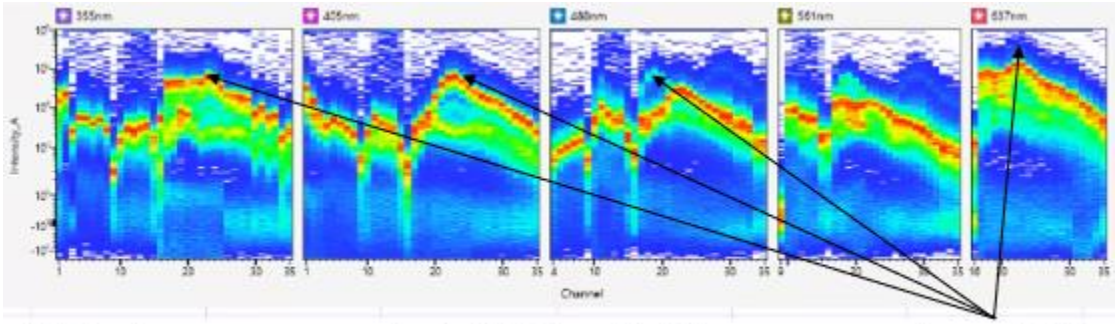
***Setting the gains for FSC, SSC and each detector array to maximize sensitivity.***

- In this next section you will use the Unstained sample to set your FSC and SSC gains and then use your brightest full stained sample to adjust the gains for each laser’s detector array. The goal is to use as high a gain as possible but still ensuring no detector within each array is off scale. Pay extra attention to the detector channels corresponding to your brightest fluorochromes.

23. Load the tube rack or plate onto the ID7000.
24. Click the “Load Tube Rack/Plate” button. 
25. Highlight the tube/well with the “Unstained” control.
26. Right click the tube/well and select “Set current position”.
27. Open the *Instrument Settings* window by clicking the “Instrument” button. 

28. Set flow rate to “1”.
29. Click the “Preview” button to begin acquiring the sample.
30. In the FCSXSSC Plot:
    a. Using the *Instruments Settings* window adjust the FSC and SSC gains to bring your cellular population/s on scale and above the threshold.
31. Highlight the tube/well with the brightest “Full Stained” sample.
32. Right click the tube/well and select “Set current position”.
33. Click the “Preview” button to begin acquiring the sample.

34. In the Spectral Ribbon plot:
    a. Using the *Instruments Settings* window, adjust the gain for each laser's detector array to ensure the brightest events in each detector array are at 10^5 or above but remain on scale (<10^6). *See example below
        i. Generally, you will want to use a gain between 4.5-5.5
        ii. Note: None of your **<u>singlet events</u>** should be above 10^6 on ANY detector.



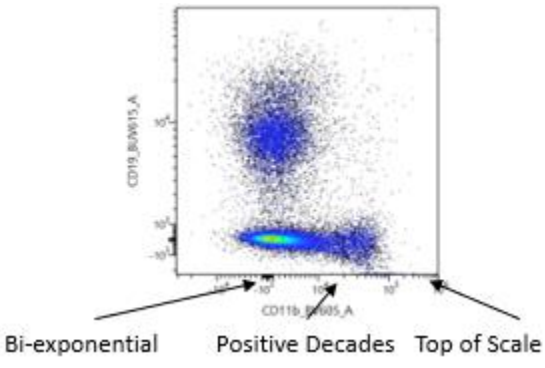
All singlet events on scale (<10^6) and brightest events above 10^5.

***Recording the experimental samples***

49. Multi-select all tubes/wells.
50. Right click and select "Add to Auto Acquisition Group"
51. Click the "Stopping Condition" button 
52. Set the events to record to desired value, be careful not to run the tube/well dry.
53. Click "Auto Acquire" button  to begin recording your tube rack/plate.
54. Click the "Unload" button  to unload the tube rack/plate.

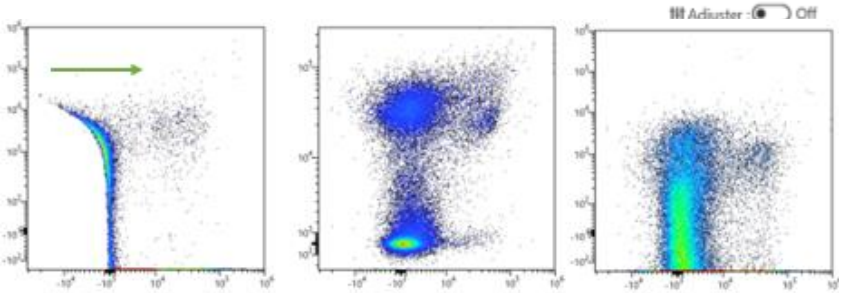
**Unmix the data and make adjustments to the unmixing**

55. Select the *Unmixing* tab at the top.
56. Click the "Unmixing Settings" button.
57. Click the "Apply" button then close the "Unmixing Settings" window.
58. Click the "Unmixing Viewer" button in the *Unmixing* tab at the top
59. Make sure the "Unmixing slider" is set to ON. 
60. Set the "Adjuster" slider to ON.
61. Fix the scaling for each parameter in your assay:
    a. Adjust the ***bi-exponential scaling***, v***isual positive decades*** and ***top of scale*** for each plot to fix visual clustering anomaly around zero, properly visualize events below/around the axis and eliminate white space in the top end . (see image above for scaling adjustment anchors)
        i. Note: It can be helpful to multi-select every plot and then click AutoXY at start.
62. Unmixing Adjustment and Correction Procedure



a. Select the current parameter in the parameter selection drop down (upper left of Unmixing viewer) - Start with the first default parameter
    i. Look through each plot and identify the population with the most egregious unmixing error.
    ii. Identify the opposing parameter on the plot containing this population.
    iii. Select the opposing parameter in the parameter selection drop down (upper left of Unmixing viewer)
    iv. Look through each plot and identify the population with the most egregious unmixing error.
    v. If the opposing parameter would send you to a new parameter return to step (i.), else if the opposing parameter would send you to the previously viewed parameter continue.
b. Set the "Adjuster" slider to ON. 

c. Drag over/under compensated population to adjust the unmixing.
    i. This can be very touchy, make small ½ pixel adjustments first.
    ii. Populations should appear rounded and generally straight. If a diagonal is witnessed, adjustment may be necessary (this means there is still correlation between the two markers in that particular plot, this could be an unmixing error or a biological phenomenon; talk to flow core personnel to discuss the particulars of your assay).




d. Create a save point by clicking the "Adjuster" slider to OFF.
e. Return to step 62a making sure to return the parameter selector to the current parameter you are correcting.
    i. Note: The first parameter you work on will take the longest, subsequent parameters will be faster.

63. After all parameters have been checked and adjusted click the "Adjuster" slider to OFF.
64. Name and save your final unmixing matrix.

**Export FCS Files**

- **It is recommended that you export your FCS unmixed (compensated) instead of raw. This is especially critical if you used controls from your Reference Library. If you instead ran a fresh set of compensation controls the day of your experiment you can export the data in a raw format (much larger file sizes), making sure to include the raw compensation data files, and unmix in**

**FlowJo. If you opt for this approach, please speak to core personnel about the procedure to unmix in FlowJo.**

65. Right click on the "Sample Group" you want to export and select "Export to FCS File" in the menu.
66. In the *FCS File Export* window:
    a. Check the checkbox for all sample groups you wish to export.
    b. Deselect the "Height" and "Width" check boxes *note: in some rare cases you may want to keep these, ask flow core personal.
    c. Select 3.1 from the "FCS version" drop down
    d. Click the "Browse" button to direct your FCS file export to your labs export folder (C:\Users\fcm\desktop\FSC Export***Your PI's last name + Lab***
    e. Click the "Export" button.
    f. Once the progress bar is finished your export will begin:
        i. The actual export occurs OUTSIDE the Sony ID7000 software and can take awhile. To track the progress please check in the bottom right side of the "Task Bar/Start Menu" near the clock and click "^", then mouse over the FCS Export applet.

**Part 5: System Clean and Shutdown**

**System Clean**

- **Complete this cleaning after each experiment.**

67. Click the *Cytometer* tab at the top.
68. Click the "Decontamination" button.
69. Select the "Bleach Cleaning and Rinse" from the drop down.
70. Follow the cleaning wizard.

**System Shutdown**

- **Only run this if you are the last person of the day.**

71. At the top of the screen select the "Cytometer" tab.
72. In the "Cytometer" tab:
    a. Click "Hardware Shutdown"
    b. Select Bleach Cleaning and Rinse.
    c. Follow the Wizard

**Estimated system flow rate by setting***

- **Note: These values are estimates and will change with each QC. Please allow a substantial buffer (>20%) to ensure wells/tubes do not run dry.**

| **Setting** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** |
|---|---|---|---|---|---|---|---|---|---|---|
| **uL/Min** | **33** | **52** | **70** | **87** | **107** | **126** | **144** | **163** | **181** | **200** |